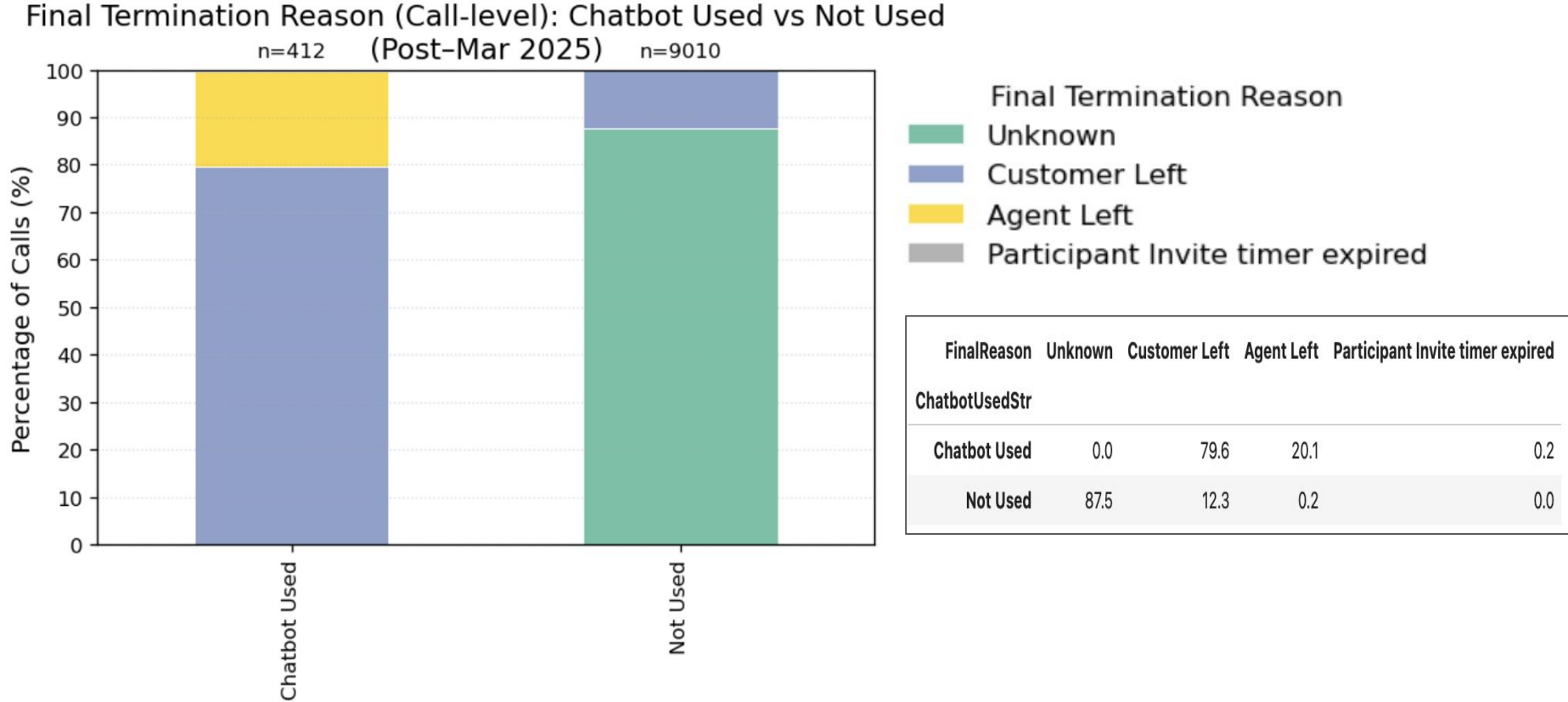


Simple EDA Team 1

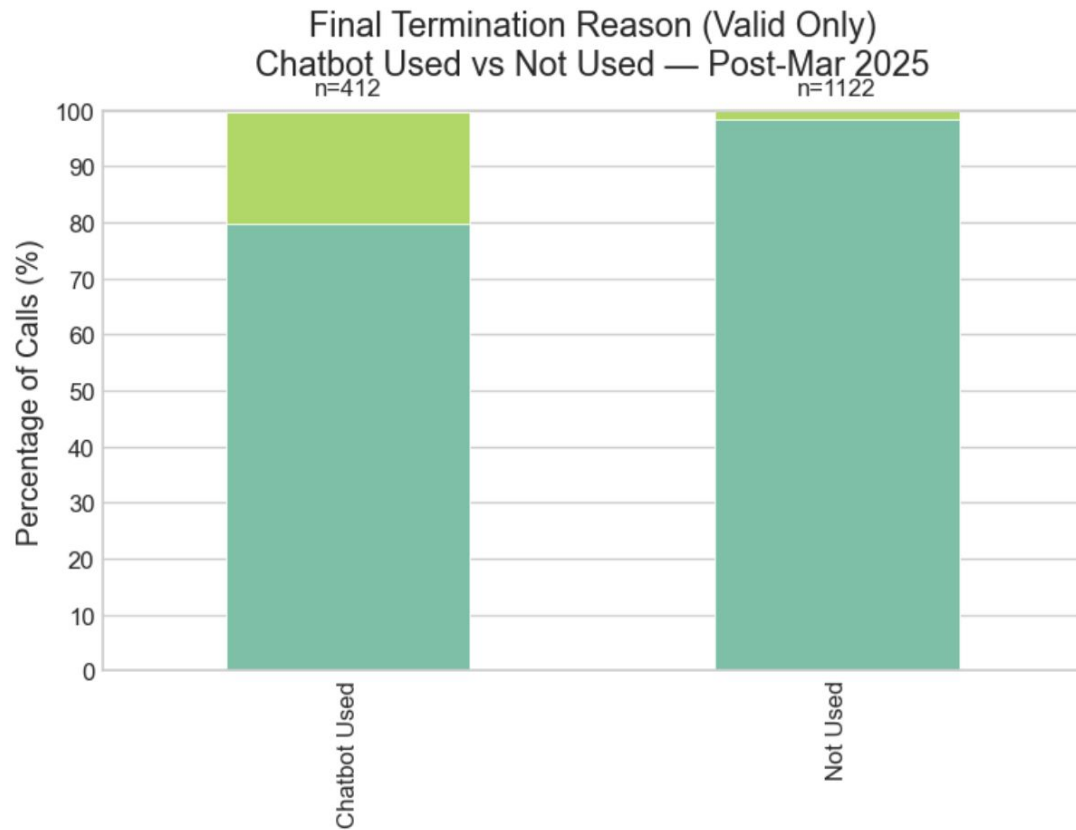
Presentation 1

Meera, Adam, Sana, Marcos

Termination Reason Analysis

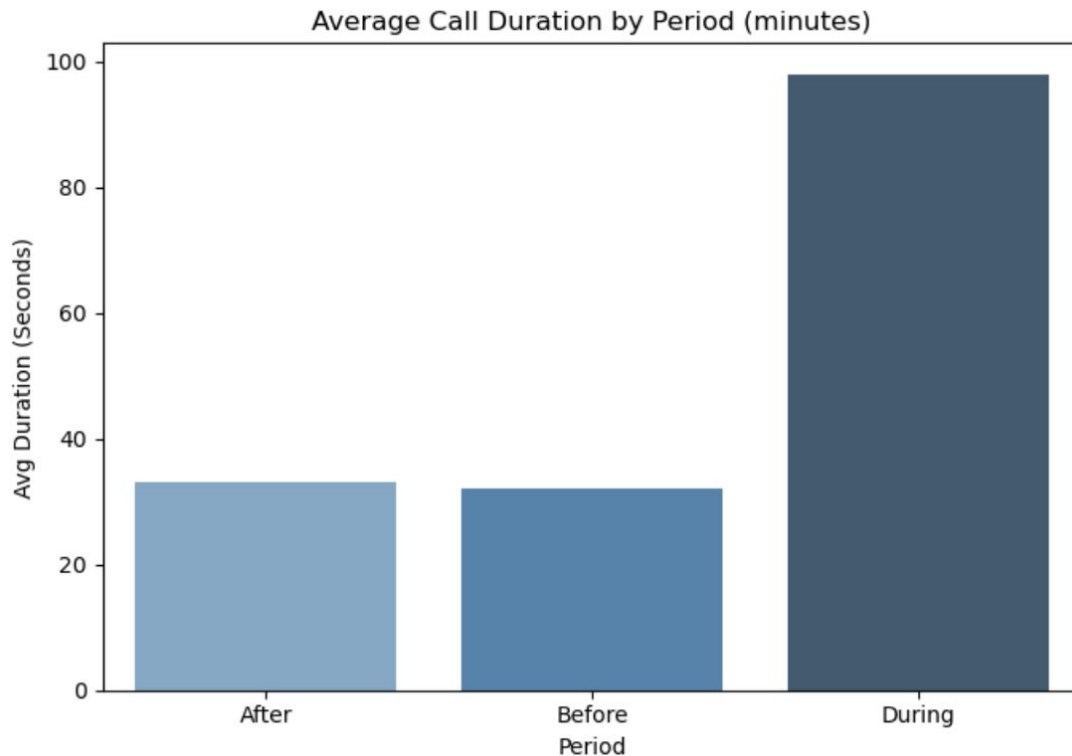


Termination Reason Analysis (w/o Unknown)



Termination Reason			
	Customer Left	Agent Left	Participant Invite timer expired
ChatbotUsedStr			
Chatbot Used	79.6	20.1	0.2
Not Used	98.4	1.6	0.0

Duration of Calls Analysis



On average call duration tripled in length during the period when the chatbot was used

Call Journey Verbiage Analysis - Meera

Purpose: find redundancies in call journey sub-menus

legal_menu_option	menu_selection	queue_selection	group_suboption	first_queue_name	final_queue_name
ADAPT	All Other Issues	ADAPT	ADAPT	ADAPT	ADAPT
ADAPT	All Other Issues	ADAPT	ADAPT	N/A	N/A
ADAPT	Housing Other	ADAPT	ADAPT	N/A	N/A
ADAPT	N/A	ADAPT	ADAPT	ADAPT	ADAPT
ADAPT	N/A	ADAPT	ADAPT	ADAPT SP	ADAPT SP
ADAPT	N/A	ADAPT	ADAPT	Clinic Voicemail Transfer	Clinic Voicemail Transfer
ADAPT	N/A	ADAPT	ADAPT	N/A	N/A
Benefits	All Other Issues	Benefits	DHS	N/A	N/A
Benefits	All Other Issues	Benefits	Other	N/A	N/A
Benefits	All Other Issues	Benefits	SSA	Benefits	Benefits

Legal Menu Option [🔗](#)

Redundancies: Legal and Other Legal?

legal_menu_option	sum_calls
ADAPT	97
Benefits	343
Consumer	464
Criminal Records	132
Employment	263
Family	1397
HIV	79
Housing	1027
Immigration	116
Legal	849
Other Legal	756
Pre-Legal Seniors	855

legal_menu_option	menu_selection	sum_calls
ADAPT	All Other Issues	11
ADAPT	Housing Other	2
Benefits	All Other Issues	7
Consumer	All Other Issues	9
Criminal Records	All Other Issues	2
Criminal Records	Personal Injury	1
Employment	Workers Comp	11
Employment	All Other Issues	2
Family	Child Support	70
Family	Simple Divorce	55
Family	All Other Issues	13
Family	From Simple Divorce to Clinics	13
Family	Criminal	1
Family	Immigration Other	1
HIV	All Other Issues	1
HIV	Criminal	1
HIV	Housing Other	1
HIV	Personal Injury	1
Housing	Housing Other	189
Housing	From Housing to Clinics	51
Housing	All Other Issues	13
Housing	Criminal	1
Immigration	Immigration Other	37

Menu Selection

Why are there separate queues for the same transfer?

- Housing >> **From Housing to Clinics >> Clinic Voicemail Transfer**
- Family >> **From Simple Divorce to Clinics >> Clinic Voicemail Transfer**
- Other Legal >> **Other Legal Issues Clinic >> Clinic Voicemail Transfer**

Immigration	All Other Issues	1
Legal	All Other Issues	24
Legal	Housing Other	12
Legal	Child Support	4
Legal	Criminal	2
Legal	Personal Injury	1
Other Legal	All Other Issues	390
Other Legal	Other Legal Issues - Clinic	141
Other Legal	Personal Injury	105
Other Legal	Criminal	88

Queue Selection

Is each subsequent column a different stage of the call? If so, why do people go through these same categories at multiple stages?

- ADAPT
- Consumer
- Education
- All Subsenior Categories

queue_selection	group_suboption	sum_calls
ADAPT	ADAPT	59
Benefits	DHS	30
Benefits	Other	9
Benefits	SSA	42
Consumer	Consumer	126
Consumer	Foreclosure	76
Consumer	Utilities	16
Education	Education	13
Employment	Other Employment	49
Employment	Unemployment	14

SubSenior ADAPT	SubSenior ADAPT	17
SubSenior Benefits	SubSenior Benefits	43
SubSenior Consumer	SubSenior Consumer	56
SubSenior Employment	SubSenior Employment	19
SubSenior Family	SubSenior Family	37
SubSenior Homeowner	SubSenior Homeowner	49
SubSenior Other	SubSenior Other	105
SubSenior Tenant	SubSenior Tenant	52

Subseniors

Redundancies: Title switch between `group_suboption` and `first_queue_name`.

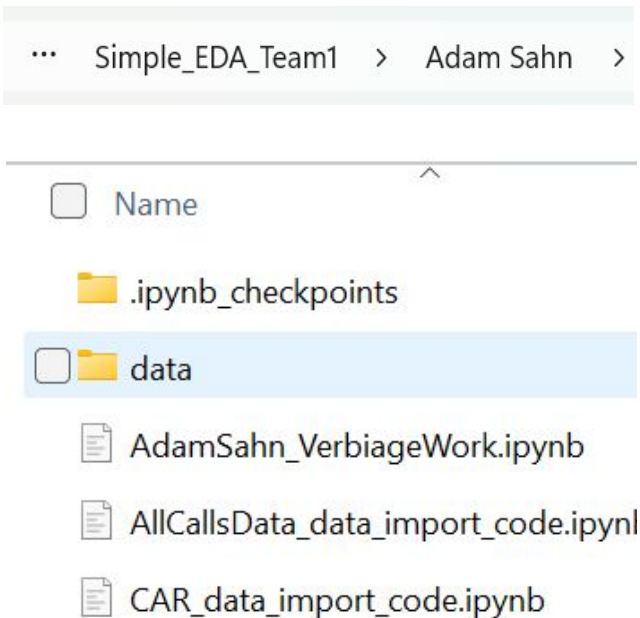
queue_selection	group_suboption	first_queue_name	final_queue_name	total_calls
SubSenior ADAPT	SubSenior ADAPT	ADAPT SubSeniors	ADAPT SubSeniors	1
SubSenior ADAPT	SubSenior ADAPT	SubSenior ADAPT	SubSenior ADAPT	15
SubSenior Benefits	SubSenior Benefits	Benefits SubSeniors	Benefits SubSeniors	4
SubSenior Benefits	SubSenior Benefits	Benefits SubSeniors SP	Benefits SubSeniors SP	1
SubSenior Benefits	SubSenior Benefits	SubSenior Benefits	SubSenior Benefits	35
SubSenior Consumer	SubSenior Consumer	Consumer SubSeniors	Consumer SubSeniors	5
SubSenior Consumer	SubSenior Consumer	SubSenior Consumer	SubSenior Consumer	36
SubSenior Consumer	SubSenior Consumer	SubSenior Consumer SP	SubSenior Consumer SP	7
SubSenior Employment	SubSenior Employment	Employment SubSeniors	Employment SubSeniors	1
SubSenior Employment	SubSenior Employment	SubSenior Employment	SubSenior Employment	13

Next Steps - Meera

1. Merge all monthly sheets into one data file (at least 2025)
2. **Research** the given information to:
 - a. Understand the difference between queue selection and group suboption
 - b. Understand the different labels (ex: SubSeniors ADAPT vs ADAPT SubSeniors)
 - i. Potentially dive deeper into optimizing this flow depending on results
3. Raise the question of **unifying the three menu selections** that all directed to Clinic Voicemail (to Cynthia)
4. Continue studying the “Correct Legal Summary Menu Reports” for **more redundancies** and open call flow doc
5. Collaborate with Adam + incorporate feedback to **refine main analysis points**

Adam Updates

Data and file organization!



Initial Verbiage Ideas

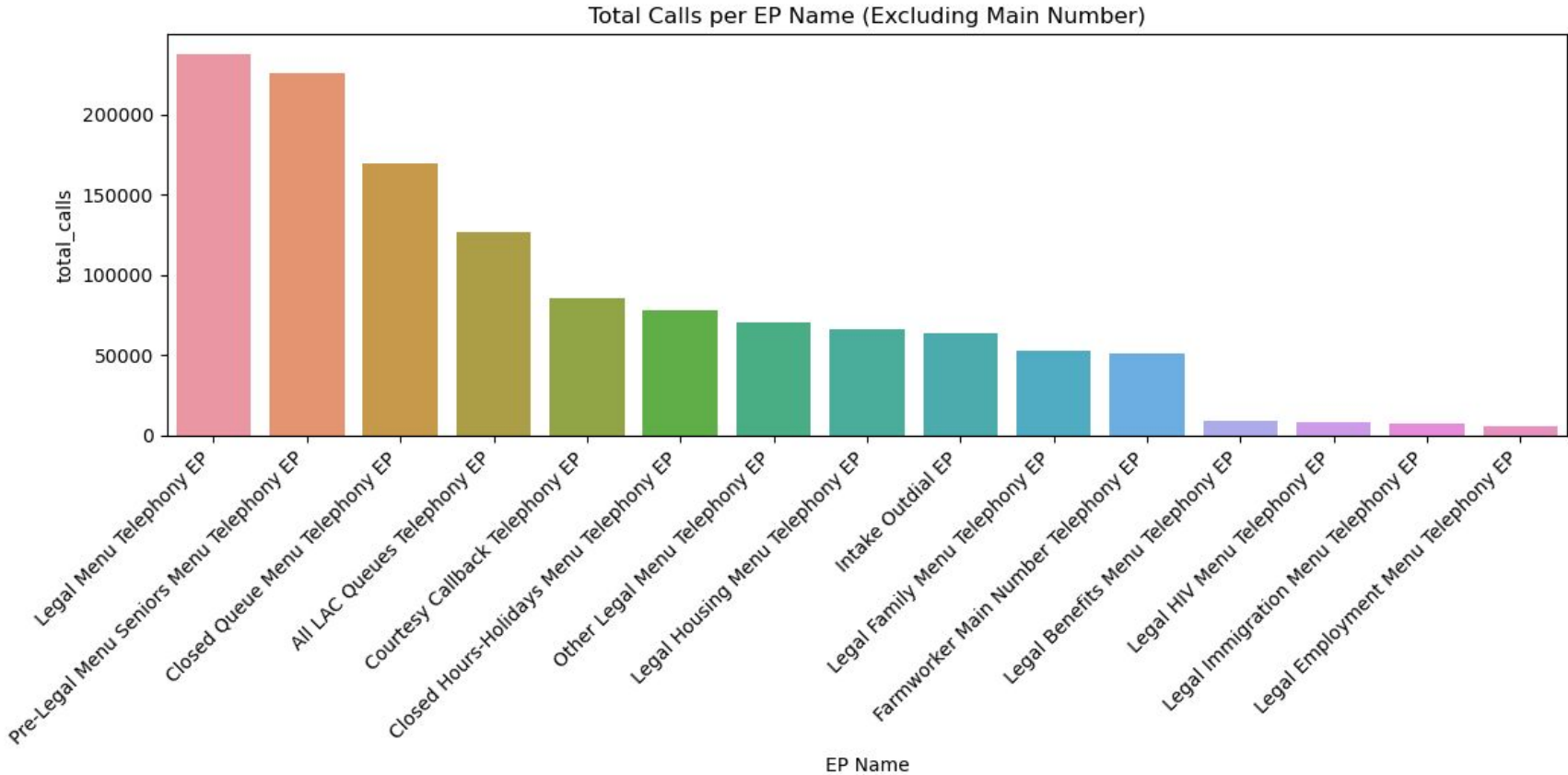
- “Thank you for calling Legal Aid Chicago. Para continuar en español, presione 2. To continue in English, press 1”
 - *Efficient if the majority of callers speak Spanish, but not if there are more English than Spanish speakers!*
- “If you are calling for help with a legal problem that you have not talked to us about yet, press 3. If you would like to hear about how to meet with a free lawyer in one of our neighborhood clinics, press 4”
 - *Wouldn't people who should be routed through #4 be likely to click #3 after hearing the prompt of 3?*

Next Steps

1. Take and implement any feedback
2. Test my verbiage ideas/hypotheses with the given data to either confirm or reject my current hypotheses (Initial Verbiage Ideas)
3. Continue looking through the “Legal Aid Chicago Contact Center Prompt List Verbiage” to identify potentially confusing or inefficient verbiage, design data-backed methods to test my hypotheses, and make actionable recommendations to Cynthia

Looking at Popular End Points from CAR Data - Sana

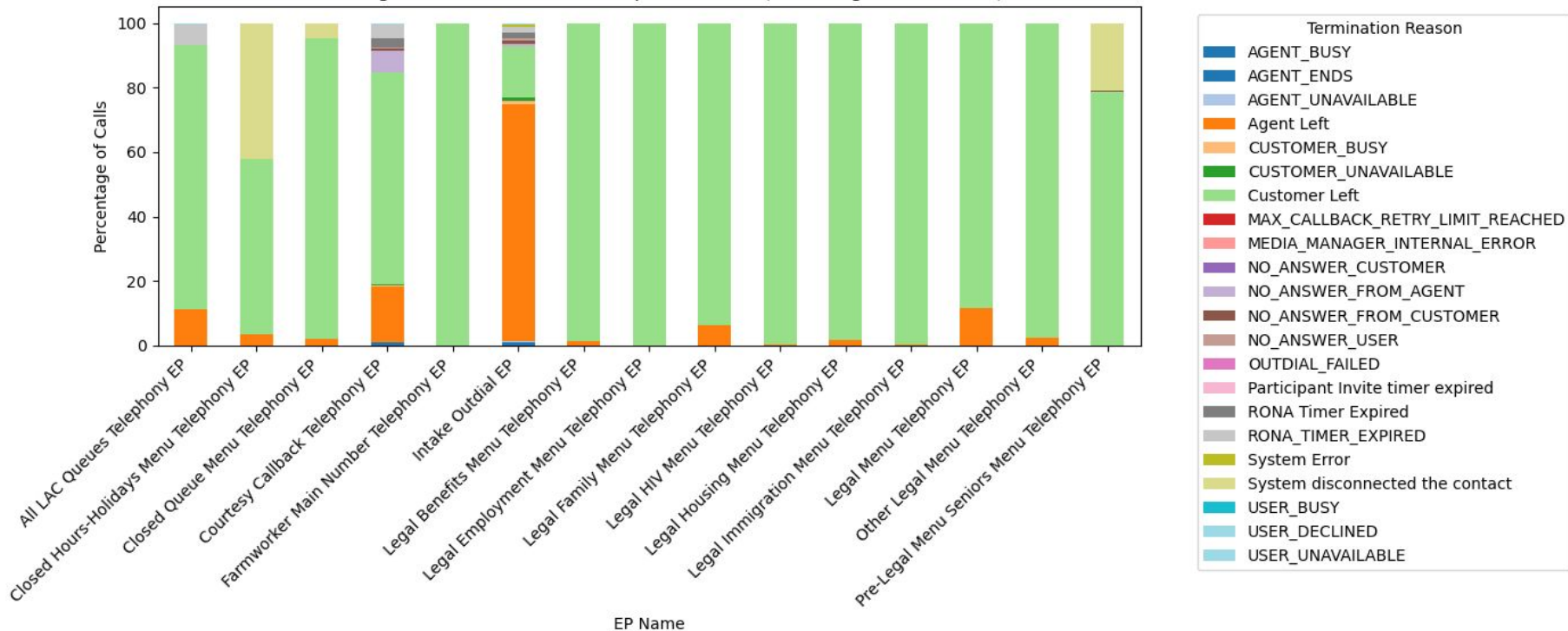
- Excluding Main Number Telephony EP (most popular)



Resource Allocation: Why are calls terminating?

- Which lines have a lot of abandoned calls → maybe under-resourced?
- System errors common?

Percentage of Termination Reasons per EP Name (Excluding Main Number)



Resource Allocation: Further interests?

- Calls by the Hour/Day of the week → M-F 8am-5pm (usual business hours)
- General focus on how resources are being utilized currently (language menu)

N/A	2025/01/13 02:46:53 PM	N/A	N/A
N/A	2025/01/13 02:46:53 PM	N/A	N/A
LanguageSelectionMenu	2025/01/13 02:46:53 PM	N/A	N/A
N/A	2025/01/13 02:46:53 PM	N/A	N/A
MainMenu	2025/01/13 02:47:01 PM	N/A	N/A
HelpWithLegalorOtherRe	2025/01/13 02:48:20 PM	N/A	N/A
MainMenu	2025/01/13 02:48:30 PM	N/A	N/A
StaffDirectorySpanishTrar	2025/01/13 02:48:58 PM	N/A	N/A
N/A	2025/01/13 02:48:58 PM	Staff Directory S	N/A
N/A	2025/01/13 02:48:58 PM	N/A	N/A
N/A	2025/01/13 02:48:59 PM	Staff Directory S	CBT Agent
N/A	2025/01/13 02:48:59 PM	Staff Directory S	CBT Agent
N/A	2025/01/13 02:50:43 PM	Staff Directory S	CBT Agent

Looking further into Spanish and English Calls

Identified two phone numbers that would lead to the English and Spanish Main Menu, filtered inbound calls, and found call duration summaries. **Spanish calls are on average 78 seconds longer than English calls.**

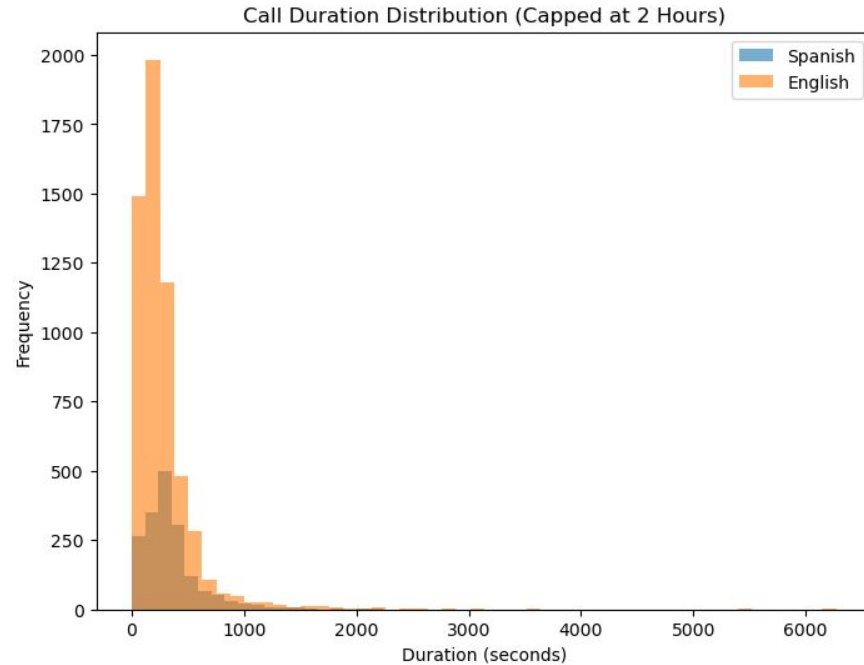
Spanish session summary:

count	1769.000000
mean	362.037875
std	350.253412
min	0.000000
25%	183.000000
50%	296.000000
75%	427.000000
max	5895.000000

English session summary:

count	5776.000000
mean	284.076524
std	366.600039
min	0.000000
25%	122.000000
50%	213.000000
75%	335.000000
max	6271.000000

General Distribution of Call Duration



Some calls were filtered (with a cap at 2 hours duration)

